

AXI4-Lite Firewall IP Core

User Guide

IP core	<code>altera_axi4_lite_firewall</code>
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1. About the AXI4-Lite Firewall IP Core

The AXI4-Lite Firewall IP core is a bump-in-the-wire component that sits between a bus master and a peripheral you want to protect. It enforces an allow-list on every transaction and guarantees the master always receives a response, even when the protected peripheral stops answering.

There is no equivalent core in the Intel FPGA IP catalog. AMD/Xilinx ship an AXI Protocol Firewall, but that core polices protocol legality and has no address-range access control; the two are complementary rather than equivalent. See [Section 9](#) for a detailed comparison.

The core is delivered as synthesisable SystemVerilog with a Platform Designer component description, a self-checking testbench, SystemVerilog assertions, and both a Questa and a licence-free Verilator simulation flow.

1.1 Features

- **Address-range access control.** Up to 64 software-programmable address ranges, each with independent read and write permission. Default-deny: a transaction matching no rule is rejected.
- **Fault isolation.** A programmable round-trip timeout on every forwarded transaction. On expiry the core answers the master itself, so a wedged peripheral cannot hang the bus master.
- **Separate control port.** Configuration and status live on a physically distinct AXI4-Lite slave, so recovery is never blocked by the fault being recovered from.
- **Interrupt output.** Level interrupt, asserted while any enabled sticky fault bit is set.
- **Isolation control.** Software can isolate the peripheral manually, and the core can isolate it automatically on a timeout.
- **Full status visibility.** Outstanding-transaction and stuck-command status bits let a driver sequence recovery correctly.
- **Secure by default.** Access control is enabled and the rule table is empty out of reset, so an unconfigured core denies everything rather than passing everything.

1.2 Device Family Support

The core is written in device-independent synthesisable SystemVerilog. It instantiates no family-specific primitives, no vendor macros, no PLLs and no memory blocks — the rule table is implemented in registers. It should therefore compile for any Intel FPGA family supported by your Quartus Prime installation.

Table 1. Device Family Support

Device family	Support level
All Intel FPGA families	Expected to compile; not characterised

Note

No device family has been characterised. The core has not been compiled by Quartus Prime at the time of writing — it has been verified only by simulation and static elaboration. See [Section 8](#) for exactly what has and has not been established.

1.3 Resource Utilization

Characterised on Intel MAX 10 by the DE10-Lite example designs in [example/](#), which instantiate the core inside complete, timing-closed systems that have been run on physical hardware.

Table 2. Resource Utilization

Item	Logic elements	Dedicated registers
<code>axi_firewall_top</code> , <code>ADDR_WIDTH</code> = 32	1,869	768
of which <code>axi_firewall_regs</code>	1,624	618
<code>axi_firewall_top</code> , <code>ADDR_WIDTH</code> = 12	1,391	544
of which <code>axi_firewall_regs</code>	1,050	327

Conditions: Intel MAX 10 10M50DAF484C7G (speed grade 7), Quartus Prime 18.1.1 Standard Edition, `DATA_WIDTH` = 32, `CTRL_ADDR_WIDTH` = 12, `NUM_RULES` = 8, `TIMEOUT_WIDTH` = 20. The core infers no memory blocks, multipliers or PLLs.

The two pairs of rows differ only in `ADDR_WIDTH`, and the core is 26 % smaller at 12 bits than at 32. That is direct evidence for the area analysis below: the rule table and its comparators dominate, and both scale with the address width.

f_{MAX} on the slow 1200 mV 85 °C model is **59.0 MHz** at `ADDR_WIDTH` = 32 (in a system closed at 50 MHz, +3.052 ns of setup slack) and **112.57 MHz** at `ADDR_WIDTH` = 12 (closed at 100 MHz, +1.117 ns). The 100 MHz result needs Quartus's High Performance Effort and physical synthesis; at default settings that design reaches 95.27 MHz and misses by 0.496 ns.

The critical path lies inside the core and is the one predicted from the source: `captured_awaddr` to `wr_timeout_cnt`, which is the combinational rule lookup feeding the forward-or-reject decision that gates the timeout counter.

The two structures that dominate area and timing are predictable from the source, and both are now confirmed by measurement:

- **The rule table** is `NUM_RULES` × (2 × `ADDR_WIDTH` + 3) registers. At the default `NUM_RULES` = 8 and `ADDR_WIDTH` = 32 that is 536 registers, which accounts for most of the 618 measured in `axi_firewall_regs`; the remainder are the control, status, interrupt and fault-capture registers.
- **The rule lookup** is a purely combinational priority chain over `NUM_RULES` entries, duplicated for the read and write paths. Each entry is two `ADDR_WIDTH`-wide magnitude comparisons. This is the measured critical path and the term that scales with `NUM_RULES`.

Note

These are measurements at two address widths on one device and speed grade, at a single value of `NUM_RULES`. f_{MAX} as a function of `NUM_RULES`, and behaviour on any other device family, remain uncharacterised. If the lookup limits f_{MAX} in your system, the standard remedy is to register the lookup result and accept one extra cycle of latency in the EVAL state.

1.4 Release Information

Table 3. Release Information

Item	Description
Version	2.0
CORE_INFO version field	0x0200
Release date	August 2026
Ordering code	None — source-available
Vendor	monkstein88
IP catalog name	AXI4-Lite Firewall
IP catalog group	Bridges and Adapters / Custom
Language	SystemVerilog (IEEE 1800)

Caution

Version 2.0 is a breaking change from 1.x. The peripheral reset output was removed and the recovery procedure changed. A 1.x driver run against a 2.0 core will acknowledge a fault and then see every subsequent transaction return SLVERR. Read the version field in `CORE_INFO` before assuming either behaviour. See [Section 7.5](#).

2. Getting Started

2.1 Installing the IP Core

The core is distributed as source. To make it visible to Platform Designer:

1. Clone or copy the repository to your machine.
2. In Quartus Prime, click **Tools ► Options ► IP Catalog Search Locations**.
3. Add the repository's **top-level directory** — the one containing `altera_axi4_lite_firewall/` — to the search path.
4. In Platform Designer, click **File ► Refresh System**.

The core appears in the IP Catalog as **AXI4-Lite Firewall** under *Bridges and Adapters / Custom*.

Note

`_hw.tcl` syntax has changed across Quartus Prime releases, and between Standard and Pro editions. The supplied `axi_firewall_hw.tcl` has not been validated against any specific release. If the component fails to import, see [Section 2.1.1](#).

2.1.1 If the Component Does Not Import

Package the component manually. This takes a few minutes and produces a `_hw.tcl` guaranteed correct for your installed toolchain:

1. In Platform Designer, click **File ► New Component**.
2. On the **Files** tab, add `rtl/axi_firewall_regs.sv` and `rtl/axi_firewall_top.sv` as synthesis files. Set `axi_firewall_top.sv` as the top-level file.
3. Click **Analyze Synthesis Files**.
4. On the **Signals & Interfaces** tab, confirm the grouping. Every port follows the `s_axi_* / m_axi_* / s_axi_ctrl_*` convention with standard AXI4-Lite suffixes, so signal analysis should infer three AXI4-Lite interfaces plus clock, reset and interrupt automatically. Correct anything it misses.
5. Click **Finish** and save.

Caution

Declare the RTL as **SystemVerilog**, not Verilog. The source uses `logic`, `always_ff`, packed structs and enumerated types. Analysed with the Verilog-2001 parser it fails on the first `logic` declaration.

2.2 Adding the Core to a Platform Designer System

Instantiate the core between the master and the peripheral to be protected:

- Connect the master's data master to `s_axi` and to `s_axi_ctrl`.
- Connect `m_axi` to the protected peripheral.
- Connect `irq` to an interrupt input on the CPU.
- Connect `clock` and `reset` to the system clock and reset.

No explicit bridge is required between Avalon-MM and AXI4-Lite. Platform Designer's interconnect provides the bridging logic. An *AXI Bridge Intel FPGA IP* may be instantiated explicitly if you want to trade concurrency for reduced interconnect logic, but it is optional.

Assign `s_axi_ctrl` a base address in the CPU's address map. The rule addresses programmed into the core refer to addresses as seen on `s_axi`, which is the protected peripheral's address decode.

2.3 Required Connections

Table 4. Connection Requirements

Connection	Requirement	Consequence if omitted
<code>s_axi</code> to the master	Required	Core does nothing
<code>m_axi</code> to the peripheral	Required	Core does nothing
<code>s_axi_ctrl</code> to a master	Required	Rule table cannot be programmed; core denies all traffic
<code>irq</code> to a CPU interrupt	Recommended	Faults must be polled instead
Software-controllable reset on the protected peripheral	Required for recovery	Cannot recover from a timeout without a full system reset

Caution

The core does not drive the protected peripheral's reset. It did in version 1.x. Recovery from a timeout still requires that the peripheral be reset, so the reset must be reachable from software — a Platform Designer reset bridge under software control, or a PIO output, both work. Without it a timeout is unrecoverable short of resetting the whole system. See [Section 3.5](#).

2.4 Simulating the IP Core

Three flows are supported. All three run the same testbench.

Verilator — no licence required, recommended for continuous integration:

```
simulation/verilator/run_sim.sh
```

The script runs a strict elaboration check, lints the RTL, builds, and runs the regression. Its exit status is 0 only if every check passes.

Questa — adds functional coverage and assertion non-vacuity reporting:

```
cd simulation/questa
vsim -c -do run_sim.tcl
```

Writes `coverage.ucdb`, `coverage_report.txt` and `run.log`. The assertion and cover-directive results are inside `coverage_report.txt`, under the bound SVA instance.

Icarus Verilog — functional tests only; the assertion bind is skipped:

```
iverilog -g2012 -DICARUS -o tb.out \
    rtl/axi_firewall_regs.sv rtl/axi_firewall_top.sv tb/axi_firewall_tb.sv
vvp tb.out
```

Note

Use `-g2012`. The sources are SystemVerilog.

2.5 Files Provided

Table 5. Files Provided

File	Description
rtl/axi_firewall_top.sv	Top level: datapaths, timeout, isolation, block/unblock
rtl/axi_firewall_regs.sv	Register block: rule table, status, interrupt, control slave
axi_firewall_hw.tcl	Platform Designer component description
tb/axi_firewall_tb.sv	Self-checking testbench, 103 checks
tb/axi_firewall_sva.sv	SystemVerilog assertions and cover points
verification/ orphan_response_tb.sv	Standalone measurement of the recovery hazard
simulation/questa/ run_sim.tcl	Questa flow with coverage
simulation/verilator/ run_sim.sh	Verilator flow
simulation/verilator/ slangcheck.py	Strict elaboration gate
example/de10_lite_rtl/	DE10-Lite example design: sixteen self-checking scenarios in synthesisable RTL, a board-level testbench, a Quartus project, and a version 2.0 C driver. See its README.md
doc/	Block diagrams and this user guide

3. Functional Description

3.1 Block Diagram

The core presents three AXI4-Lite interfaces and one interrupt output.

Traffic flows from the master into `s_axi`, is checked, and is either forwarded to the peripheral through `m_axi` or answered locally with an error response. Configuration and status are reached through `s_axi_ctrl`, which is independent of the data path — so configuring or inspecting the core is never subject to a firewall rule, and never blockable by a peripheral that has stopped answering. Figure 1 shows how the core wires into a system.

Internally the core contains two independent datapaths — one for writes, one for reads — and a shared register block. Each datapath has its own state machine, its own capture registers, its own rule-lookup port and its own fault signals, so a read and a write may be in flight simultaneously. The only shared state is the fault capture pair `FAULT_ADDR` and `FAULT_INFO`. Figure 2 shows the internal structure.

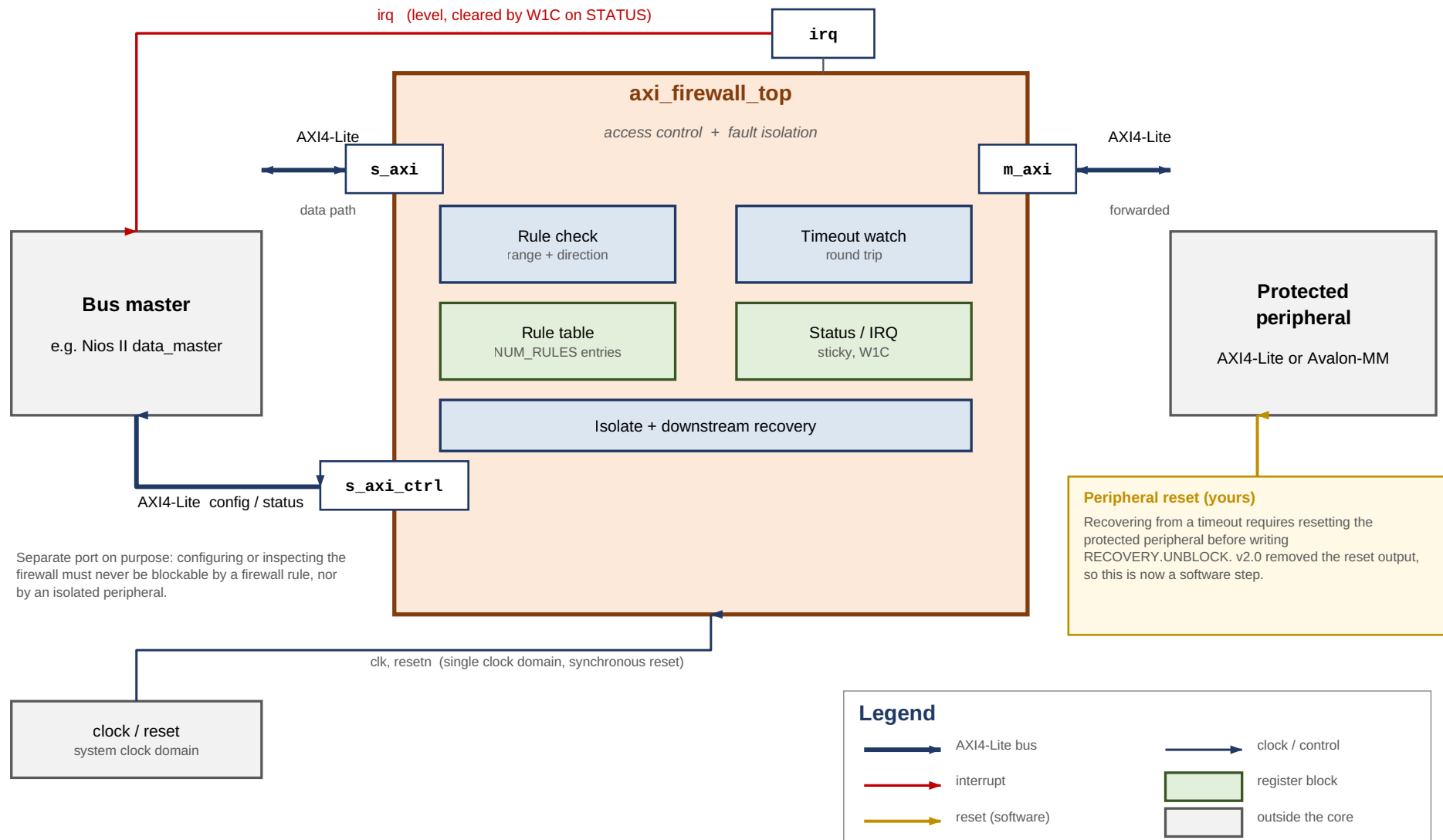


Figure 1. System Context

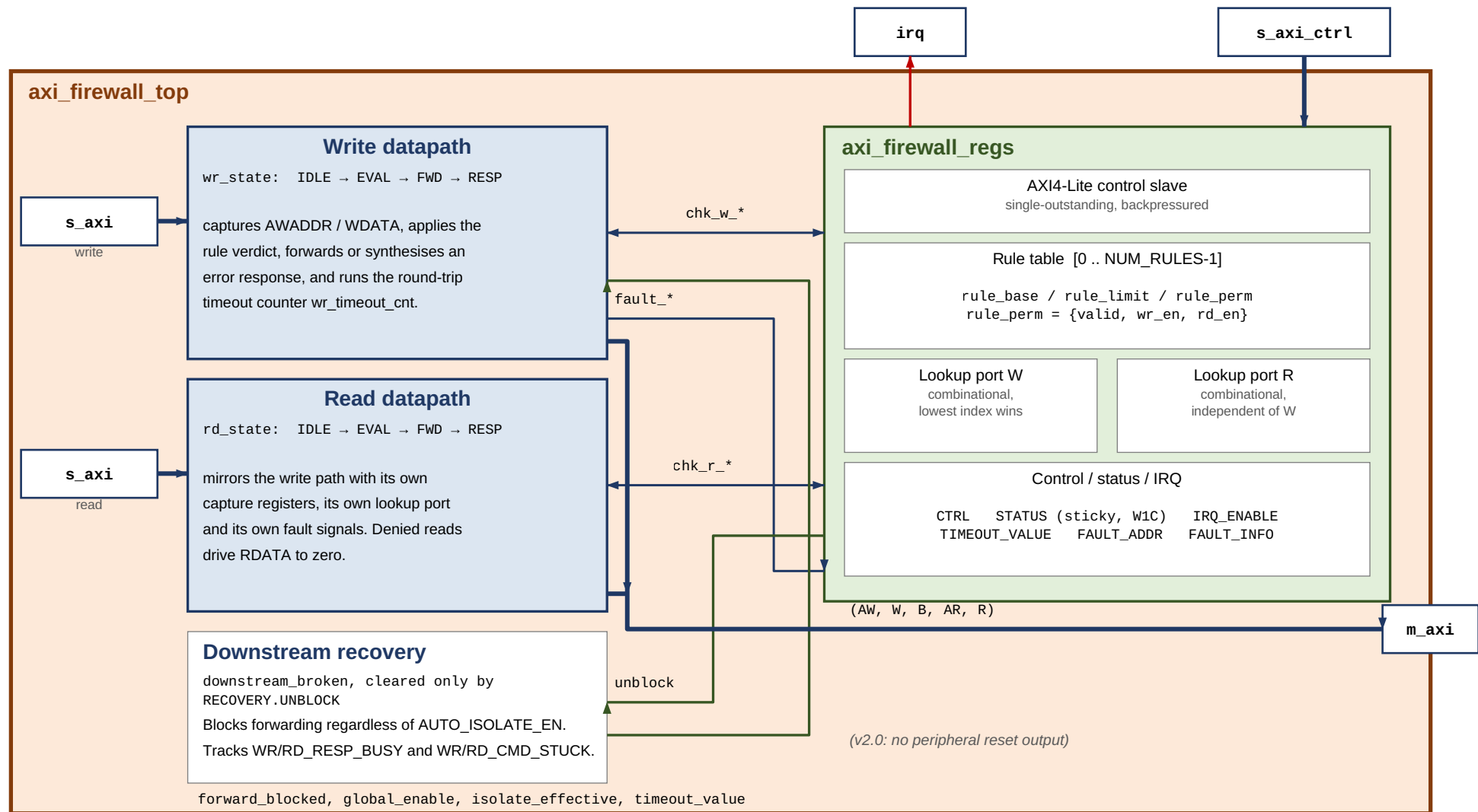


Figure 2. Internal Architecture

3.2 Access Control

Every transaction's address is compared against the rule table. Each rule is an inclusive address range with independent read and write permission, and a valid bit.

The verdict is applied in the EVAL state, in strict priority order:

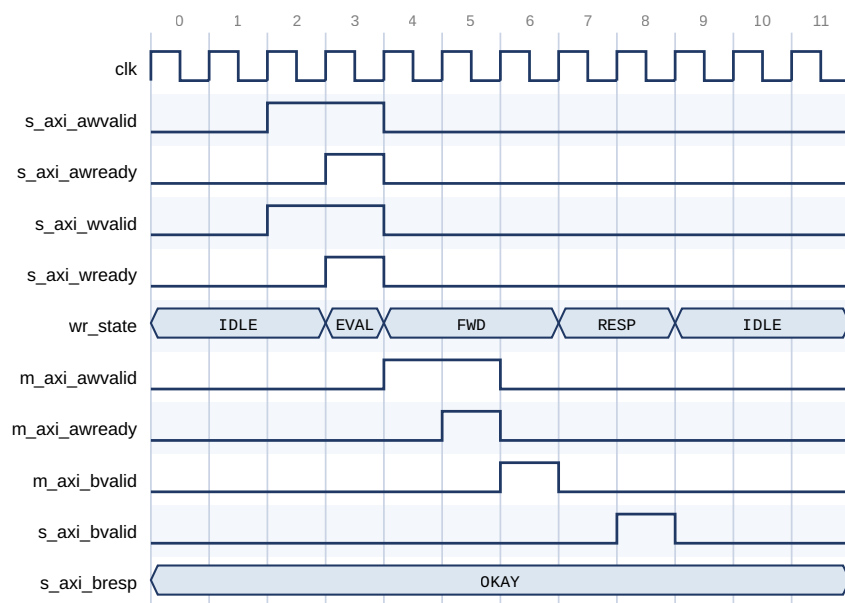
1. If the downstream is blocked or the core is isolated, respond SLVERR.
2. If `CTRL.GLOBAL_ENABLE` is clear, forward unconditionally (bypass mode).
3. If no valid rule contains the address, respond DECERR.
4. If a rule contains the address but does not permit this direction, respond SLVERR.
5. Otherwise, forward the transaction.

Table 6. Response Encoding

Condition	Response	Status bit set	Interrupt
Permitted, peripheral answers	As returned by the peripheral	—	No
No valid rule matches	DECERR	ADDR_VIOLATION	Yes, if enabled
Rule matches, direction denied	SLVERR	PERM_VIOLATION	Yes, if enabled
Blocked or isolated	SLVERR	None	No
Peripheral timed out	SLVERR	TIMEOUT_ERROR	Yes, if enabled

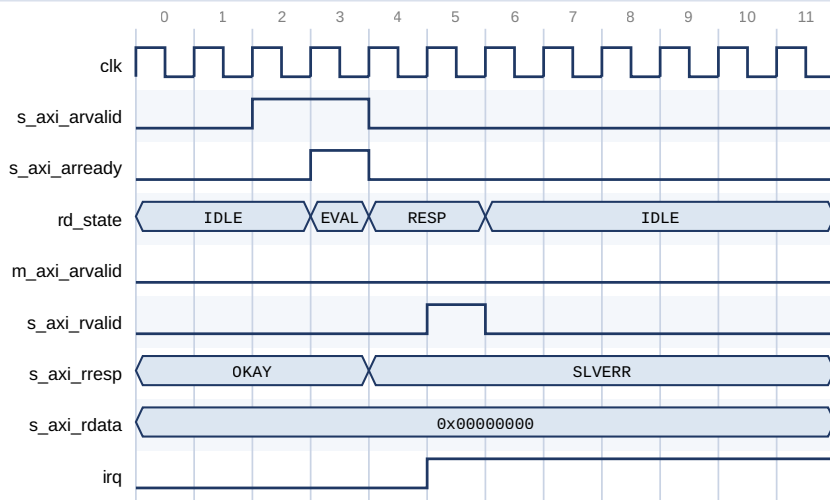
A transaction rejected because the core is isolated or blocked sets no status bit and raises no interrupt. This is deliberate: only genuine policy violations and timeouts are logged, so a burst of rejected traffic during isolation cannot bury the fault that caused it.

A denied read drives `RDATA` to zero rather than leaving the previous value on the bus, so a rejected read cannot leak the result of an earlier permitted one.



The address matches a valid rule that permits writes, so the transaction is forwarded on `m_axi`.
Cycle 1: request accepted (AWREADY and WREADY together). Cycle 6: BVALID returns OKAY.

Figure 3. Permitted Write — Request to Response, 6 Cycles



The address falls in a write-only rule. RD_EVAL rejects it and answers SLVERR from RD_RESP.
 m_axi_arvalid never asserts - the peripheral is not touched. RDATA is driven to zero, not left stale.
 irq asserts because STATUS.PERM_VIOLATION is sticky and its interrupt is enabled.

Figure 4. Permission-Denied Read — Answered Locally, Nothing Reaches the Peripheral

In Figure 4, note that `m_axi_arvalid` never asserts. The peripheral is not touched at all, `RDATA` is driven to zero, and `irq` asserts because the sticky `PERM_VIOLATION` bit was set with its interrupt enabled.

3.3 Rule Evaluation

The lookup is a combinational priority chain: the **lowest-index valid rule** containing the address wins. Ranges need not be disjoint — if you use overlapping ranges, place the more specific rule at the lower index.

Two independent lookup ports are provided, one per datapath, so a read and a write are resolved in the same cycle without contending for a shared comparator.

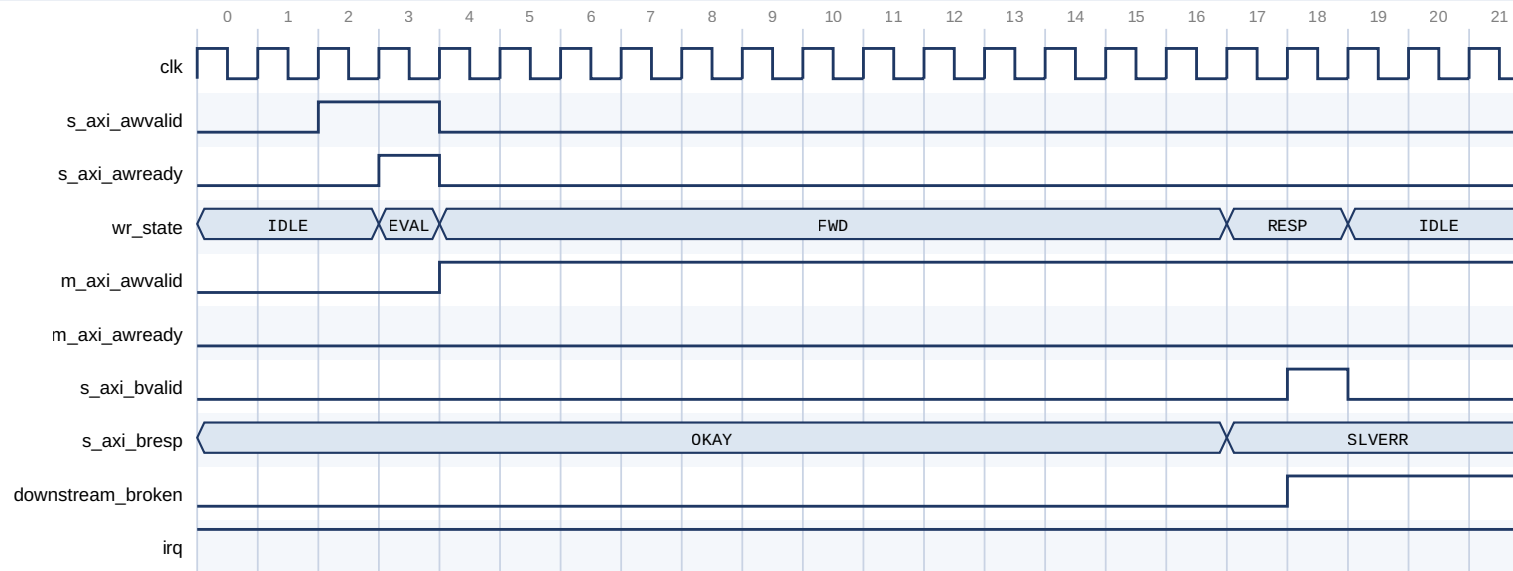
The lookup is purely combinational, so the verdict costs no extra cycle: EVAL is always one cycle.

3.4 Fault Isolation and Timeout

Every forwarded transaction is watched by a counter loaded from `TIMEOUT_VALUE`. The counter covers the **entire round trip** — from the moment forwarding begins to the arrival of the response — so it catches both a peripheral that never accepts the command and one that accepts it and then goes quiet.

On expiry the core:

1. answers **SLVERR upstream immediately**, so the master never hangs;
2. latches an internal *downstream-blocked* state that blocks all further forwarding, independently of `CTRL.AUTO_ISOLATE_EN`;
3. **leaves any asserted `m_axi_*VALID` asserted**, because AXI requires `VALID` to remain asserted until `READY`.



The peripheral never raises AWREADY. TIMEOUT_VALUE is 12 clocks in this capture.

On expiry the core answers SLVERR upstream so the master never hangs, and latches downstream_broken.

m_axi_awvalid stays asserted: AXI forbids withdrawing VALID before READY. Only UNBLOCK may drop it.

Figure 5. Downstream Timeout — Master Answered Immediately, VALID Left Asserted

Caution

Withdrawing an asserted `VALID` before its handshake is an AXI protocol violation that can wedge the interconnect between the core and the peripheral, not merely the peripheral. The core therefore never does so on timeout. The stuck `VALID` is withdrawn at exactly one point — the `RECOVERY.UNBLOCK` write described in [Section 3.5](#).

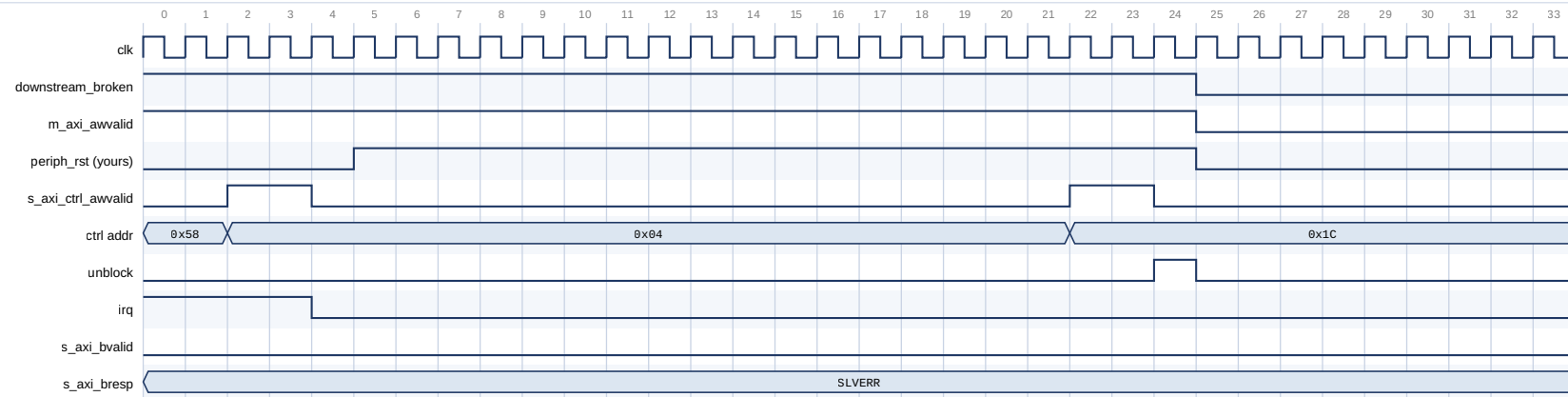
`CTRL.AUTO_ISOLATE_EN` governs only whether the visible `ISOLATED` status bit is set. Blocking after a timeout is required for protocol safety and happens either way.

3.5 Recovery Sequence

Recovery is an explicit software sequence. It is modelled on the procedure AMD document for their AXI Firewall, which carries the same requirement to reset the monitored side before unblocking.

Table 7. Recovery Sequence

Step	Action	Why
1	Stop issuing transactions to <code>s_axi</code>	New transactions are rejected while blocked
2	Write 1 to the sticky <code>STATUS</code> bits	Acknowledges the fault and releases the auto-isolate latch. Required before step 5 — see below
3	Poll <code>STATUS</code> until <code>WR_RESP_BUSY</code> and <code>RD_RESP_BUSY</code> clear — with a bound	Tells you no late response is still in flight
4	Assert the protected peripheral's reset and hold it (≥ 16 clocks)	Discards the peripheral's AXI state
5	Write 1 to <code>RECOVERY.UNBLOCK</code> , with the peripheral still held in reset	Reopens forwarding and withdraws the stuck <code>VALID</code> . Holding reset means nothing downstream can latch the command as it is withdrawn — see below
6	Release the peripheral's reset	The peripheral restarts with no knowledge of the abandoned transaction
7	Resume, retrying anything that failed	Transactions attempted while blocked returned <code>SLVERR</code>



Write 0x04 acknowledges the fault (irq drops). The peripheral reset is driven by the system, not the core.
 Write 0x1C pulses unblock: downstream_broken clears and the stuck m_axi_awvalid is withdrawn in the same event.
 The final write completes with OKAY, showing forwarding has reopened.

Figure 6. Recovery Sequence — Acknowledge, Reset the Peripheral, UNBLOCK

Caution

Step 4 is not optional. `UNBLOCK` causes the core to withdraw an asserted `VALID`. If the peripheral has not been reset, that is a protocol violation on a live bus, and the peripheral may additionally still act on a transaction the core already reported to the master as failed. Measured: 0 of 25 tested timing offsets mis-attribute a stale response when the sequence is followed; 1 of 25 does when step 4 is skipped.

Caution

The reset must still be asserted when step 5 executes. Reset is a held state across steps 4 to 6, not a pulse to be fired and forgotten before unblocking.

The reason is that the orphaned command is still on `m_axi` until `UNBLOCK` retracts it. A peripheral that has been reset and *released* sees a perfectly valid `AWVALID`/`ARVALID` waiting on the bus and accepts it, committing a transaction the master was already told had failed. Unlike the stale-response hazard above, which is timing-dependent and was measured at 1 offset in 25, this one is deterministic: it happens every time.

Scenarios `b` and `C` of the DE10-Lite example design are this sequence with the two orderings, and read back `0x00000000` and `0xCAFEF00D` respectively from the protected peripheral.

Caution

Step 2 must precede step 5, and the order is load-bearing. Forwarding is gated by two independent conditions — the blocked state and the isolate state — and each is cleared by a different write. `UNBLOCK` clears only the blocked state. If `AUTO_ISOLATE_EN` was set when the timeout occurred, the auto-isolate latch is also set, and only the `W1C` in step 2 releases it. Writing `UNBLOCK` without having acknowledged `STATUS` first leaves the core isolated, and every transaction still returns `SLVERR`. `STATUS` then reads `BLOCKED = 0` with `ISOLATED = 1`, which is the signature of this mistake.

3.5.1 Bounding the Busy Poll

`WR_RESP_BUSY` and `RD_RESP_BUSY` mean *the peripheral has accepted a command and still owes a response*. A peripheral that accepted a command and then died owes one forever, so an unbounded poll in step 3 hangs precisely when recovery matters most.

Treat the bits as advisory:

- **Clear** — no late response can still be in flight; the reset is unambiguously safe.
- **Stuck** — reset anyway, and let `UNBLOCK` discard what is owed.

`WR_CMD_STUCK` and `RD_CMD_STUCK` report the other case: a command the peripheral never accepted at all. These can only be cleared by `UNBLOCK`, so a driver that sees one knows immediately that polling `RESP_BUSY` will not be enough.

Note

This differs from AMD's core, which autonomously flushes outstanding transactions when blocked so that its busy bits always reach zero. Removing this caveat by adding autonomous flushing is the most valuable planned enhancement to this core.

3.6 Interrupts

`irq` is a level interrupt, asserted while any sticky `STATUS` bit is set and its corresponding `IRQ_ENABLE` bit is set:

```
irq = (ADDR_VIOLATION & IRQ_ENABLE[0])
      | (PERM_VIOLATION & IRQ_ENABLE[1])
      | (TIMEOUT_ERROR & IRQ_ENABLE[2])
```

Clear the interrupt at its source by writing 1 to the relevant `STATUS` bit. There is no separate acknowledge register. This is the standard memory-mapped-peripheral idiom and works directly with the Nios II HAL ISR pattern.

`FAULT_ADDR` and `FAULT_INFO` capture the address and type of the most recent fault of any kind.

Caution

If the read and write datapaths fault in the *same cycle* with *different* fault types, the capture registers can disagree with each other. `FAULT_ADDR` and `FAULT_INFO.WAS_WRITE` always take the write side. `FAULT_INFO.FAULT_TYPE` is resolved separately, by type precedence across both datapaths — `TIMEOUT` beats `PERM` beats `ADDR`. So a simultaneous write-`ADDR` and read-`PERM` fault reports `WAS_WRITE = 1` with the write's address, but `FAULT_TYPE = PERM`, which belongs to the read.

The sticky `STATUS` bits are unaffected — every fault sets its own bit correctly. Only the single shared capture pair is ambiguous. Software that must attribute every fault individually should correlate with its own transaction log rather than rely on `FAULT_INFO` alone; see [Section 9.7](#).

3.7 Control Port Behaviour

`s_axi_ctrl` is single-outstanding and applies backpressure accordingly: `AWREADY` and `WREADY` are withheld while a `BVALID` is unacknowledged, and `ARREADY` while an `RVALID` is. A master that issues one access at a time sees no difference; one that pipelines simply waits.

Writes to reserved or unaligned offsets are ignored and answered `OKAY`; reads of them return zero.

The control port is deliberately independent of the data path. Configuring or inspecting the core is never subject to a firewall rule, and never blockable by an isolated or hung peripheral — which is what makes recovery possible at all.

3.8 Reset

`resetn` is active-low and **synchronous**. It resets the entire core: both datapath state machines, the rule table, all status and configuration registers, and the blocked state.

A global reset is therefore also the escape hatch of last resort for a downstream that never recovers — it clears the blocked state without an `UNBLOCK`, at the cost of losing the rule table, which must be reprogrammed.

Out of reset the core is **secure by default**: `GLOBAL_ENABLE` is 1 and the rule table is empty, so every transaction is denied until rules are programmed.

3.9 Latency

Table 8. Latency, Zero-Wait-State Peripheral

Operation	Cycles
Single write, request asserted to <code>BVALID</code>	6
Single read, request asserted to <code>RVALID</code>	6

Measured in simulation against a zero-wait-state peripheral model, counting clock edges from request assertion to response valid. The testbench fails the run if either exceeds 8 cycles, so the figures cannot silently regress.

The core is single-outstanding and non-pipelined, so this cost is per transaction and does not amortise.

4. Parameters

All five parameters are compile-time. Nothing in the core is runtime-reconfigurable except the rule table and the control registers.

Table 9. Parameters

Parameter	Type	Default	Legal range	Description
ADDR_WIDTH	Integer	32	8–32	Width of <code>s_axi / m_axi</code> address buses. Also the width of every rule base and limit, and of <code>FAULT_ADDR</code> .
DATA_WIDTH	Integer	32	32, 64	Width of <code>s_axi / m_axi</code> data buses. <code>WSTRB</code> is <code>DATA_WIDTH/8</code> bits.
CTRL_ADDR_WIDTH	Integer	12	8–16	Width of the <code>s_axi_ctrl</code> address bus.
NUM_RULES	Integer	8	1–64	Number of address-range rules.
TIMEOUT_WIDTH	Integer	20	8–32	Width of the timeout counter. Maximum programmable timeout is $2^{\text{TIMEOUT_WIDTH}} - 1$ clock cycles.

Ranges are enforced by `axi_firewall_hw.tcl`. `DATA_WIDTH` is restricted to 32 and 64 because AXI4-Lite defines only those two.

Sizing CTRL_ADDR_WIDTH

The register file spans `0x40 + NUM_RULES × 16` bytes, so:

$$\text{CTRL_ADDR_WIDTH} \geq \text{ceil}(\log_2(0x40 + \text{NUM_RULES} \times 16))$$

Table 10. Minimum CTRL_ADDR_WIDTH

NUM_RULES	Table spans	Highest byte used	Minimum CTRL_ADDR_WIDTH
1	0x50 bytes	0x4F	7
4	0x80 bytes	0x7F	7
8 (default)	0xC0 bytes	0xBF	8
16	0x140 bytes	0x13F	9
32	0x240 bytes	0x23F	10
64	0x440 bytes	0x43F	11

The default of 12 covers every legal `NUM_RULES`. A validation callback in `axi_firewall_hw.tcl` rejects an undersized value, because the failure mode is silent: high-index rules simply become unreachable, the rule table looks like it programmed correctly, and the firewall quietly enforces fewer rules than you configured.

Sizing TIMEOUT_WIDTH

Choose the timeout to be comfortably longer than the protected peripheral's worst-case response, including any interconnect arbitration. `TIMEOUT_WIDTH` only has to be wide enough to express it.

Table 11. Maximum Timeout by Width

TIMEOUT_WIDTH	Maximum count	At 100 MHz	At 50 MHz
8	255	2.6 μ s	5.1 μ s
12	4 095	41 μ s	82 μ s
16	65 535	655 μ s	1.3 ms
20 (default)	1 048 575	10.5 ms	21 ms
24	16 777 215	168 ms	336 ms
32	4 294 967 295	43 s	86 s

Note

TIMEOUT_VALUE resets to all ones, which at the default width is 10.5 ms at 100 MHz — long enough that the timeout is effectively disabled until software programs a real value. This is deliberate: the reset state denies traffic on policy grounds, but never manufactures a spurious timeout during boot.

5. Interface Signals

The core has 60 ports across six Platform Designer interfaces.

Table 12. Interfaces

Interface	Type	Role	Signals
clock	Clock sink	Single clock domain	1
reset	Reset sink	Active-low, synchronous	1
s_axi	AXI4-Lite slave	Protected data path, master side	19
m_axi	AXI4-Lite master	Protected data path, peripheral side	19
s_axi_ctrl	AXI4-Lite slave	Configuration and status	19
irq	Interrupt sender	Fault interrupt	1

5.1 Clock and Reset

Table 13. Clock and Reset Signals

Signal	Direction	Width	Description
clk	Input	1	Clock. All three interfaces are synchronous to it; there is no CDC in the core.
resetn	Input	1	Active-low synchronous reset. Resets the FSMs, rule table and all registers.

5.2 s_axi — Protected Slave

Connect to the bus master. Standard AXI4-Lite slave.

Table 14. s_axi Signals

Signal	Direction	Width	Description
s_axi_awaddr	Input	ADDR_WIDTH	Write address
s_axi_awprot	Input	3	Protection type. Accepted and forwarded unchanged; not used in rule evaluation.
s_axi_awvalid	Input	1	Write address valid
s_axi_awready	Output	1	Write address ready
s_axi_wdata	Input	DATA_WIDTH	Write data
s_axi_wstrb	Input	DATA_WIDTH/8	Write strobes
s_axi_wvalid	Input	1	Write data valid
s_axi_wready	Output	1	Write data ready
s_axi_bresp	Output	2	Write response: OKAY , SLVERR or DECERR
s_axi_bvalid	Output	1	Write response valid
s_axi_bready	Input	1	Write response ready
s_axi_araddr	Input	ADDR_WIDTH	Read address
s_axi_arprot	Input	3	Protection type. Forwarded unchanged; not used in rule evaluation.
s_axi_arvalid	Input	1	Read address valid
s_axi_arready	Output	1	Read address ready
s_axi_rdata	Output	DATA_WIDTH	Read data. Driven to zero on any denied read.
s_axi_rresp	Output	2	Read response
s_axi_rvalid	Output	1	Read data valid
s_axi_rready	Input	1	Read data ready

Note

AWPROT / ARPROT are transported but not policed. The rule table keys on address and direction only. If you need privilege-based filtering, that is a feature addition, not a configuration option.

5.3 m_axi — Protected Master

Connect to the peripheral being protected. Signal names and widths mirror s_axi with directions reversed.

Table 15. m_axi Signals of Note

Signal	Direction	Width	Description
m_axi_awaddr	Output	ADDR_WIDTH	Forwarded write address, unmodified
m_axi_awvalid	Output	1	Held asserted until AWREADY , including across a timeout
m_axi_wvalid	Output	1	Held asserted until WREADY , including across a timeout
m_axi_bready	Output	1	Tied high. The core always accepts a write response.
m_axi_rready	Output	1	Tied high. The core always accepts read data.

Caution

`m_axi_bready` and `m_axi_rready` are tied high so that a peripheral can never be back-pressured into a deadlock by the firewall itself. The consequence is that a late response from an abandoned transaction is accepted rather than rejected, which is exactly why the recovery sequence resets the peripheral before unblocking. See [Section 3.5](#).

5.4 `s_axi_ctrl` — Control and Status Slave

A separate AXI4-Lite slave carrying the register file. `WDATA` is always 32 bits wide and `WSTRB` always 4 bits, independent of `DATA_WIDTH`; the addressable range is set by `CTRL_ADDR_WIDTH`.

Table 16. `s_axi_ctrl` Signals of Note

Signal	Direction	Width	Description
<code>s_axi_ctrl_awaddr</code>	Input	<code>CTRL_ADDR_WIDTH</code>	Register byte offset
<code>s_axi_ctrl_wdata</code>	Input	32	Always 32 bits
<code>s_axi_ctrl_wstrb</code>	Input	4	Byte strobes; honoured for partial-width writes
<code>s_axi_ctrl_bresp</code>	Output	2	Always <code>OKAY</code> ; writes to reserved offsets are ignored, not errored
<code>s_axi_ctrl_rdata</code>	Output	32	Reserved offsets read as zero

5.5 `irq`

Table 17. `irq`

Signal	Direction	Width	Description
<code>irq</code>	Output	1	Active-high level interrupt. Asserted while any enabled sticky <code>STATUS</code> bit is set. Cleared by <code>W1C</code> on <code>STATUS</code> .

6. Register Map

All registers are 32 bits and word-aligned on `s_axi_ctrl`. Offsets are byte offsets from the base address assigned to `s_axi_ctrl`.

Table 18. Register Map

Offset	Name	Access	Reset	Description
0x00	CTRL	R/W	0x3	Global enable, auto-isolate, manual isolate
0x04	STATUS	R / W1C	0x0	Sticky faults, live isolation and outstanding status
0x08	IRQ_ENABLE	R/W	0x7	Per-fault interrupt enables
0x0C	TIMEOUT_VALUE	R/W	all ones	Round-trip timeout, in clock cycles
0x10	FAULT_ADDR	R	0x0	Address of the most recent fault
0x14	FAULT_INFO	R	0x0	Type and direction of the most recent fault
0x18	CORE_INFO	R	—	Version and NUM_RULES
0x1C	RECOVERY	W	—	UNBLOCK . Reads as zero.
0x20–0x3F	<i>reserved</i>	—	—	Writes ignored, reads return zero
0x40 + i·0x10	RULE_BASE[i]	R/W	0x0	Inclusive range base
0x44 + i·0x10	RULE_LIMIT[i]	R/W	0x0	Inclusive range limit
0x48 + i·0x10	RULE_PERM[i]	R/W	0x0	Permissions and valid bit
0x4C + i·0x10	<i>reserved</i>	—	—	Padding to a 16-byte stride

i runs 0 to NUM_RULES – 1 . With the default 8 rules the table spans 0x40–0xBF.

6.1 CTRL (0x00)

Table 19. CTRL

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	MANUAL_ISOLATE	R/W	0	1 = reject all traffic with SLVERR , regardless of the rule table
1	AUTO_ISOLATE_EN	R/W	1	1 = a timeout also sets the visible ISOLATED status
0	GLOBAL_ENABLE	R/W	1	1 = enforce the rule table; 0 = forward everything unchecked

Caution

Clearing GLOBAL_ENABLE disables access control entirely — the core becomes a transparent pass-through with only the timeout still active. It is a bring-up and debug aid, not an operating mode.

6.2 STATUS (0x04)

Table 20. STATUS

Bit	Name	Access	Reset	Description
31:9	reserved	R	0	Reads zero
8	RD_CMD_STUCK	R	0	Blocked and <code>m_axi_arvalid</code> still asserted — a read the peripheral never accepted
7	WR_CMD_STUCK	R	0	Blocked and <code>m_axi_awvalid</code> or <code>m_axi_wvalid</code> still asserted — a write the peripheral never accepted
6	RD_RESP_BUSY	R	0	The peripheral accepted a read command and still owes <code>RVALID</code>
5	WR_RESP_BUSY	R	0	The peripheral accepted a full write and still owes <code>BVALID</code>
4	BLOCKED	R	0	Forwarding is blocked after a timeout; cleared only by <code>RECOVERY.UNBLOCK</code>
3	ISOLATED	R	0	<code>MANUAL_ISOLATE</code> or the auto-isolate latch is active
2	TIMEOUT_ERROR	R/W1C	0	Sticky: a forwarded transaction timed out
1	PERM_VIOLATION	R/W1C	0	Sticky: a rule matched but denied the direction
0	ADDR_VIOLATION	R/W1C	0	Sticky: no valid rule matched the address

Bits 8:3 are live status, not sticky, and cannot be written. Bits 2:0 are write-1-to-clear; writing 0 to a bit leaves it unchanged.

Caution

Writing 1 to `TIMEOUT_ERROR` acknowledges the fault and releases the auto-isolate latch, but does **not** clear `BLOCKED` and does not resume forwarding. That requires `RECOVERY.UNBLOCK`. In version 1.x the `W1C` alone resumed traffic; the separation exists so that acknowledging a fault cannot accidentally restart traffic toward a peripheral nobody has reset.

6.3 IRQ_ENABLE (0x08)

Table 21. IRQ_ENABLE

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	EN_TIMEOUT	R/W	1	Enable irq on <code>TIMEOUT_ERROR</code>
1	EN_PERM	R/W	1	Enable irq on <code>PERM_VIOLATION</code>
0	EN_ADDR	R/W	1	Enable irq on <code>ADDR_VIOLATION</code>

Masking a bit suppresses the interrupt only. The corresponding `STATUS` bit is still set and still requires a `W1C` to clear.

6.4 TIMEOUT_VALUE (0x0C)

Table 22. TIMEOUT_VALUE

Bit	Name	Access	Reset	Description
31:TIMEOUT_WIDTH	reserved	R	0	Reads zero; writes to these bits are discarded
TIMEOUT_WIDTH-1:0	TIMEOUT	R/W	all ones	Round-trip timeout in clock cycles

Byte strobes are honoured, so a byte- or halfword-wide write updates only the addressed bytes of the field. Programming 0 makes every forwarded transaction time out immediately; the core does not reject the value.

6.5 FAULT_ADDR (0x10)

Table 23. FAULT_ADDR

Bit	Name	Access	Reset	Description
31:ADDR_WIDTH	reserved	R	0	Reads zero when ADDR_WIDTH < 32
ADDR_WIDTH-1:0	ADDR	R	0	Address of the most recent fault

Overwritten by every new fault, including one whose interrupt is masked. Read it before clearing STATUS if you intend to log it.

6.6 FAULT_INFO (0x14)

Table 24. FAULT_INFO

Bit	Name	Access	Reset	Description
31:4	reserved	R	0	Reads zero
3:1	FAULT_TYPE	R	0	See table below
0	WAS_WRITE	R	0	1 = the faulting transaction was a write

Table 25. FAULT_TYPE Encoding

Value	Name	Meaning
0	NONE	No fault recorded since reset
1	ADDR	No valid rule matched — answered DECERR
2	PERM	A rule matched but denied the direction — answered SLVERR
3	TIMEOUT	The peripheral did not complete in time — answered SLVERR

Values 4–7 are not generated.

6.7 CORE_INFO (0x18)

Table 26. CORE_INFO

Bit	Name	Access	Description
31:16	VERSION	R	0x0200 for version 2.0
15:8	reserved	R	Reads zero
7:0	NUM_RULES	R	Rule count this instance was built with

Read NUM_RULES at run time rather than hard-coding it, so one driver binary works across instances. Check VERSION before running the recovery sequence: 1.x and 2.0 recover differently.

6.8 RECOVERY (0x1C)

Table 27. RECOVERY

Bit	Name	Access	Description
31:1	reserved	W	Ignored
0	UNBLOCK	W1S, self-clearing	Writing 1 clears BLOCKED , reopens forwarding and withdraws any stuck m_axi VALID

Reads return zero. Writing 0 does nothing. New in version 2.0.

Caution

Only write UNBLOCK after resetting the protected peripheral. This is the one point at which the core withdraws an asserted VALID , which is safe against a peripheral that has just been reset and a protocol violation against one that has not.

6.9 Rule Table

Each rule occupies a 16-byte slot at $0x40 + i \times 0x10$.

Table 28. Rule Slot Layout

Sub-offset	Name	Access	Reset	Description
+0x0	RULE_BASE[i]	R/W	0	Inclusive base address
+0x4	RULE_LIMIT[i]	R/W	0	Inclusive limit address
+0x8	RULE_PERM[i]	R/W	0	Permissions, see below
+0xC	reserved	—	—	Padding

Table 29. RULE_PERM[i]

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	VALID	R/W	0	1 = this rule takes part in matching
1	WRITE_ALLOW	R/W	0	1 = permit writes in this range
0	READ_ALLOW	R/W	0	1 = permit reads in this range

A rule matches when VALID is set and $RULE_BASE[i] \leq \text{address} \leq RULE_LIMIT[i]$. Both bounds are inclusive, so a single-word window is $BASE == LIMIT$. $BASE > LIMIT$ matches nothing; the core does not reject it.

The lowest-index matching valid rule wins. Only its permission bits are consulted — a later, more permissive rule covering the same address has no effect.

RULE_BASE and RULE_LIMIT honour byte strobes, so a byte-wide write updates only the addressed bytes. RULE_PERM is updated only when WSTRB[0] is set.

7. Software Programming Model

Register offsets in the examples below are relative to the base address assigned to `s_axi_ctrl`.

```
#define FW_CTRL            0x00
#define FW_STATUS          0x04
#define FW_IRQ_ENABLE      0x08
#define FW_TIMEOUT         0x0C
#define FW_FAULT_ADDR      0x10
#define FW_FAULT_INFO      0x14
#define FW_CORE_INFO       0x18
#define FW_RECOVERY         0x1C
#define FW_RULE(i)         (0x40 + (i) * 0x10)
#define FW_RULE_BASE(i)    (FW_RULE(i) + 0x0)
#define FW_RULE_LIMIT(i)   (FW_RULE(i) + 0x4)
#define FW_RULE_PERM(i)    (FW_RULE(i) + 0x8)

#define FW_PERM_VALID      (1u << 2)
#define FW_PERM_WRITE      (1u << 1)
#define FW_PERM_READ       (1u << 0)

#define FW_ST_ADDR_VIOL    (1u << 0)
#define FW_ST_PERM_VIOL    (1u << 1)
#define FW_ST_TIMEOUT      (1u << 2)
#define FW_ST_ISOLATED     (1u << 3)
#define FW_ST_BLOCKED      (1u << 4)
#define FW_ST_WR_BUSY      (1u << 5)
#define FW_ST_RD_BUSY      (1u << 6)
#define FW_ST_WR_STUCK     (1u << 7)
#define FW_ST_RD_STUCK     (1u << 8)
#define FW_ST_FAULTS       (FW_ST_ADDR_VIOL | FW_ST_PERM_VIOL | FW_ST_TIMEOUT)
```

7.1 Initialisation

Program the rule table before allowing any traffic to reach `s_axi`. Out of reset the table is empty and `GLOBAL_ENABLE` is set, so every transaction is denied until you configure it.

```

void fw_init(void *base, uint32_t timeout_cycles)
{
    /* Rules are only meaningful up to the count this instance was built
       with. Read it rather than assuming the default. */
    uint32_t n_rules = IORD_32DIRECT(base, FW_CORE_INFO) & 0xFF;

    for (uint32_t i = 0; i < n_rules; i++)
        IOWR_32DIRECT(base, FW_RULE_PERM(i), 0); /* invalidate */

    IOWR_32DIRECT(base, FW_TIMEOUT, timeout_cycles);
    IOWR_32DIRECT(base, FW_STATUS, FW_ST_FAULTS); /* clear stale sticky */
    IOWR_32DIRECT(base, FW_IRQ_ENABLE, 0x7);
}

void fw_add_rule(void *base, uint32_t i,
                 uint32_t lo, uint32_t hi, int can_read, int can_write)
{
    /* Program the range before the valid bit. Setting VALID first would
       expose a window in which the rule is live over a stale range. */
    IOWR_32DIRECT(base, FW_RULE_BASE(i), lo);
    IOWR_32DIRECT(base, FW_RULE_LIMIT(i), hi);
    IOWR_32DIRECT(base, FW_RULE_PERM(i),
                  FW_PERM_VALID
                    | (can_write ? FW_PERM_WRITE : 0)
                    | (can_read ? FW_PERM_READ : 0));
}

```

Caution

Write `RULE_PERM` last. The rule table is live: a rule becomes active the instant its `VALID` bit is set, and a transaction arriving in the window between setting `VALID` and updating the range would be judged against whatever the range registers happened to hold.

7.2 Changing a Rule at Run Time

To narrow a rule safely, clear `VALID` first, change the range, then set `VALID` again. Widening a rule can be done range-first without an intermediate invalidation, because the intermediate state is never more permissive than either endpoint.

7.3 Interrupt Service Routine

```
void fw_isr(void *base)
{
    uint32_t st = IORD_32DIRECT(base, FW_STATUS);

    /* Capture before acknowledging: a later fault overwrites these. */
    uint32_t addr = IORD_32DIRECT(base, FW_FAULT_ADDR);
    uint32_t info = IORD_32DIRECT(base, FW_FAULT_INFO);

    if (st & FW_ST_ADDR_VIOL) log_addr_violation(addr, info & 1);
    if (st & FW_ST_PERM_VIOL) log_perm_violation(addr, info & 1);
    if (st & FW_ST_TIMEOUT)    schedule_recovery();

    /* W1C only the bits actually seen, so a fault raised between the read
       and the write is not silently discarded. */
    IOWR_32DIRECT(base, FW_STATUS, st & FW_ST_FAULTS);
}
```

Do not run the recovery sequence from inside the ISR. It resets the peripheral and polls, which does not belong in interrupt context.

7.4 Recovery from a Timeout

This implements the sequence in [Section 3.5](#).

Note the shape of the reset interface: **two functions, not one**. A single `reset_peripheral()` that pulses and returns cannot express this sequence, because the reset has to still be asserted when `UNBLOCK` is written.

```

int fw_recover(void *base,
               void (*reset_assert)(void), /* drive the peripheral's reset */
               void (*reset_release)(void), /* and release it */
               void (*delay_clocks)(unsigned))
{
    uint32_t st;

    /* Step 1 is the caller's: stop issuing transactions to the
       protected peripheral before calling this function. */

    /* Step 2: acknowledge the fault and release the auto-isolate latch.
       This does NOT reopen the downstream - that is step 5. */
    IOWR_32DIRECT(base, FW_STATUS, FW_ST_FAULTS);

    /* Step 3: bounded poll. A peripheral that accepted a command and then
       died owes a response forever, so this MUST NOT be unbounded. On
       expiry, continue anyway - UNBLOCK discards what is still owed. */
    for (int spin = 0; spin < 1000; spin++) {
        st = IORD_32DIRECT(base, FW_STATUS);
        if (!(st & (FW_ST_WR_BUSY | FW_ST_RD_BUSY)))
            break;
    }

    /* Step 4: assert the protected peripheral's reset and HOLD it. Not
       optional. At least 16 clocks so its AXI state machines fully
       re-initialise. */
    reset_assert();
    delay_clocks(16);

    /* Step 5: reopen the downstream, with the peripheral STILL IN RESET.
       UNBLOCK is the point at which the core withdraws the VALID left over
       from the timed-out command. Until then that command is still sitting
       on m_axi, and a peripheral that is out of reset will accept it - see
       the second Caution in Section 3.5. */
    IOWR_32DIRECT(base, FW_RECOVERY, 1);

    /* Step 6: only now release the peripheral. */
    reset_release();

    /* Step 7: the caller retries anything that failed while blocked. */
    return (IORD_32DIRECT(base, FW_STATUS) & FW_ST_BLOCKED) ? -1 : 0;
}

```

Note

A tested implementation of this function, with the reset hooks parameterised and the ordering checked by host tests, is provided as `example/de10_lite_nios/software/axi4_lite_firewall.c`.

Caution

The bound on step 3 is the important part. `WR_RESP_BUSY` and `RD_RESP_BUSY` clear when the peripheral delivers what it owes, and a dead peripheral never will. An unbounded poll hangs the recovery path in exactly the case it exists for. Check `WR_CMD_STUCK` and `RD_CMD_STUCK` too: those can only be cleared by `UNBLOCK`, so seeing either tells you immediately that polling `RESP_BUSY` will not be enough.

7.5 Migrating from Version 1.x

Table 30. Version 1.x to 2.0 Changes

Area	Version 1.x	Version 2.0
Peripheral reset	<code>m_axi_resetn</code> output driven by the core	Port removed; resetting the peripheral is a software step
<code>RESET_HOLD_CYCLES</code> parameter	Present	Removed
Resuming after a timeout	W1C on <code>STATUS.TIMEOUT_ERROR</code>	W1C then <code>RECOVERY.UNBLOCK</code>
<code>RECOVERY</code> register	Absent	New at 0x1C
<code>STATUS</code> bits 8:4	Absent	<code>BLOCKED</code> , <code>WR/RD_RESP_BUSY</code> , <code>WR/RD_CMD_STUCK</code>
<code>CORE_INFO</code> version	0x0102	0x0200

Migration steps:

1. Remove the `m_axi_resetn` connection from the Platform Designer system and provide a software-controllable reset to the peripheral instead.
2. Replace every "clear `TIMEOUT_ERROR` to resume" site with the six-step sequence in [Section 7.4](#).
3. Gate the new path on `CORE_INFO[31:16]` if one driver must support both.

A 1.x driver run unchanged against a 2.0 core does not fail loudly. It acknowledges the fault, never writes `UNBLOCK`, and every subsequent transaction returns `SLVERR` from a core that stays blocked forever.

8. Verification

The core is verified by simulation and static elaboration. This section states precisely what has been established and what has not.

8.1 Verification Environment

Three front ends are used, because each catches things the others miss.

Table 31. Verification Tools

Tool	Version	Role	What it uniquely caught
slang	11	Strict IEEE 1800 elaboration gate	Mixed <code>timescale</code> between files
Verilator	5.48	Regression and lint, 9 parameter configurations	Lint findings across the parameter sweep
Questa	2024.1	Functional and assertion coverage	Declaration-order errors Verilator accepts

The Questa finding is worth stating plainly: three constructs in the version 2.0 RTL elaborated cleanly under Verilator and were rejected by Questa. A port connection to an identifier declared later in the file creates an implicit net that then collides with the explicit declaration. Verilator tolerated it; Questa was right to refuse. That is why the toolchain has three front ends rather than one.

8.2 Results

All figures below are from the version 2.0 sources.

Table 32. Verification Results

Metric	Result
Self-checking testbench checks	103 / 103 pass
SystemVerilog assertions	14 / 14, zero failures
Assertions with a non-vacuous pass	14 / 14
Cover directives	6 / 6 hit
FSM states covered	8 / 8
FSM transitions covered	14 / 14
<code>m_axi</code> protocol violations observed	0
Total functional coverage (Questa)	85.96 %
slang errors	0
Verilator lint findings, 9 configurations	0

Table 33. Assertions

Assertion	Passes	Checks
<code>a_suppress_illegal_write</code>	4	A denied write never reaches <code>m_axi</code>
<code>a_suppress_illegal_read</code>	4	A denied read never reaches <code>m_axi</code>
<code>a_err_on_blocked_write</code>	4	A blocked write is answered with an error, not silence
<code>a_err_on_blocked_read</code>	4	Ditto for reads
<code>a_awvalid_stability</code>	23	Slave-side <code>AWVALID</code> /payload stable until <code>AWREADY</code>
<code>a_arvalid_stability</code>	23	Slave-side <code>ARVALID</code> stability
<code>a_bvalid_stability</code>	6	<code>BVALID</code> / <code>BRESP</code> stable until <code>BREADY</code>
<code>a_rvalid_stability</code>	6	<code>RVALID</code> / <code>RDATA</code> / <code>RRESP</code> stable until <code>RREADY</code>
<code>a_m_awvalid_stability</code>	162	Master-side <code>AWVALID</code> stability, excluding the <code>unblock</code> discard
<code>a_m_wvalid_stability</code>	162	Master-side <code>WVALID</code> stability
<code>a_m_arvalid_stability</code>	119	Master-side <code>ARVALID</code> stability
<code>a_no_issue_while_blocked</code>	382	No write is issued to <code>m_axi</code> while blocked
<code>a_no_read_issue_while_blocked</code>	429	No read is issued while blocked
<code>a_block_holds_until_unblock</code>	560	<code>BLOCKED</code> persists until <code>UNBLOCK</code> , never self-clears

Every assertion has a non-zero non-vacuous pass count. This matters: an assertion that only ever passes vacuously is a comment with syntax highlighting. Two assertions in an earlier revision were exactly that, and the fix was adding a test that made the antecedent true, not editing the assertion.

Table 34. Cover Directives

Directive	Hits	What it proves the suite reaches
<code>c_write_denied</code>	4	A permission-denied write actually occurs
<code>c_read_denied</code>	4	A permission-denied read actually occurs
<code>c_write_decerr</code>	3	An out-of-range write reaching <code>DECERR</code>
<code>c_read_decerr</code>	2	An out-of-range read reaching <code>DECERR</code>
<code>c_block_and_recover</code>	5	A full block-and-recover cycle
<code>c_unblock_with_stuck_cmd</code>	3	<code>UNBLOCK</code> while a command is still stuck on <code>m_axi</code>

Table 35. FSM Transition Coverage

Transition	Write FSM	Read FSM	Exercised by
IDLE → EVAL	23	23	Every transaction reaching the core
EVAL → FWD	17	16	Every permitted transaction
EVAL → RESP	5	6	Denied, out-of-range and blocked transactions
EVAL → IDLE	1	1	Reset asserted during EVAL
FWD → RESP	14	13	Normal completion and timeout
FWD → IDLE	3	3	Reset asserted during FWD
RESP → IDLE	19	19	Every completed response

8.3 The Orphan-Response Measurement

`verification/orphan_response_tb.sv` measures the hazard that motivates step 4 of the recovery sequence. It runs a clean, permitted write, injects one late `BVALID` from an abandoned transaction at offset k cycles, and sweeps k over 25 values. If the master ever sees `SLVERR` on a write the peripheral answered `OKAY`, a stale response was mis-attributed.

Table 36. Orphan-Response Results

Recovery sequence	Offsets that mis-attribute
Peripheral reset before <code>UNBLOCK</code> (documented)	0 of 25
<code>UNBLOCK</code> without resetting the peripheral	1 of 25

This is why step 4 is stated as mandatory rather than recommended.

8.4 Hardware Demonstration

The example design in `example/de10_lite_rtl/` is a second, independent verification vehicle, and covers ground the testbench structurally cannot: it is the only place the core is driven by **synthesisable hardware** rather than by testbench tasks, through the same RTL that is programmed into an FPGA.

A hardware sequencer programs the rule table over `s_axi_ctrl`, issues transactions at `s_axi`, injects faults into the protected peripheral, and checks every response against the register map in Section 6. Sixteen scenarios cover permitted and denied access in both directions, both violation types, both timeout shapes, isolation, the recovery sequence, bypass and interrupt masking.

There are two example designs, and they answer different questions. The RTL one asks whether the core behaves; the Nios II one asks whether it *integrates*, by putting it behind generated Platform Designer interconnect, an Avalon-to-AXI bridge and a processor data cache.

Table 37. Hardware Demonstration Results

Metric	RTL example	Nios II example
Driven by	a hardware sequencer	C on a Nios II/f
Clock	50 MHz	100 MHz (PLL)
<code>ADDR_WIDTH</code>	32	12
Passing on a physical DE10-Lite	16 / 16 scenarios	33 / 33 checks
Board-level checks in simulation	80 / 80, Questa and Verilator	not applicable
Quartus 18.1.1 compilation	0 errors	0 errors
Timing closure	met, +3.052 ns	met, +1.117 ns
Driver host tests	not applicable	29 / 29

Two results from that work are folded back into this document. The resource and f_{MAX} figures in Section 1.3 come from it. And the second Caution in Section 3.5 — that the peripheral's reset must still be asserted when `UNBLOCK` is written — was found by building it: scenarios **b** and **C** are the same recovery with the two orderings, and the wrong one deterministically commits a write the master was told had failed.

Both were run on a Terasic DE10-Lite. The RTL example carries an In-System Sources and Probes instance so a script can drive it and read its pass bitmap back over JTAG; the Nios II example reports over its JTAG UART.

Note

Running on hardware exposed two defects that simulation had not, and both were in the Nios II *test program* rather than in the core. An interrupt handler that acknowledged faults was clearing the sticky `STATUS` bits before the checking code could read them, and posted writes meant `STATUS` could be sampled before the offending transaction had been evaluated. Neither is reachable in a testbench, which has no interrupt latency, no write posting and no cache. They are recorded here because they are the failure modes a real driver will meet.

8.5 What Has Not Been Verified

Stated plainly, because the absence of a result is easy to mistake for a passing one.

Table 38. Not Verified

Item	Status
Quartus Prime analysis of the <code>_hw.tcl</code> component	Verified for Quartus 18.1.1 Standard. Platform Designer recognises all six interfaces and all five parameters, and the component generates and synthesises inside a real system
Synthesis results (logic elements, registers)	Measured at <code>NUM_RULES</code> = 8 on MAX 10 — see Section 1.3
Timing closure at 50 MHz on MAX 10	Met — see Section 1.3
f_{MAX} vs <code>NUM_RULES</code>	Never swept. One data point only
Behaviour inside a generated Platform Designer interconnect	Verified by the Nios II example, on hardware
Device families other than MAX 10	None characterised
Behaviour on physical hardware	Verified on a Terasic DE10-Lite (10M50DAF484C7G), both examples
Formal property proof	Not attempted; assertions are simulation-only

Everything in Sections 1 through 7 that describes core behaviour is backed by simulation, and the parts a demonstration can reach are additionally confirmed on silicon. What remains untested is breadth rather than depth: one device family, one speed grade, two address widths, one value of `NUM_RULES`, and no formal proof.

9. Design Considerations and Limitations

9.1 Single Outstanding Transaction

Each datapath handles one transaction at a time. A read and a write may overlap, but a second write waits for the first to complete. Interconnect generated by Platform Designer will pipeline requests into the firewall, which simply back-pressures.

This is a deliberate simplification. The core answers the master itself on a timeout, and doing that correctly with multiple outstanding transactions requires tracking which responses are still owed and which have been synthesised — the exact bookkeeping that `RESP_BUSY` and `CMD_STUCK` expose in rudimentary form today.

9.2 No Autonomous Flushing

When blocked, the core stops forwarding but does not itself complete or discard transactions the peripheral already accepted. The consequence is the bounded poll in [Section 3.5.1](#).

AMD's AXI Protocol Firewall does flush autonomously, which is why its busy bits always reach zero and its recovery has no equivalent caveat. Adding this is the most valuable planned enhancement.

9.3 Address-Only Policy

Rules key on address range and direction. `AWPROT / ARPROT` are forwarded but not evaluated, so the core cannot express "privileged accesses only" or "instruction fetches only".

9.4 No Protocol Legality Checking

The core does not police AXI legality on `s_axi`. A master that violates VALID stability will not be caught here. This is the main functional difference from AMD's core, which is a protocol checker with no address-based access control. The two are complementary.

Table 39. Comparison with the AMD AXI Protocol Firewall

Capability	This core	AMD AXI Protocol Firewall
Address-range access control	Yes, up to 64 rules	No
Read/write permission per range	Yes	No
AXI protocol legality checking	No	Yes, extensive
Timeout on the protected side	Yes	Yes
Guaranteed response to the master	Yes	Yes
Blocks after a fault until unblocked	Yes	Yes
Autonomous flushing when blocked	No	Yes
Requires resetting the monitored side before unblock	Yes	Yes
Full AXI4 (bursts, IDs)	No, AXI4-Lite only	Yes

9.5 Rule Table Is Not Lockable

Any master that can reach `s_axi_ctrl` can rewrite the rule table, including disabling the firewall through `GLOBAL_ENABLE`. The core protects a peripheral from a misbehaving master on `s_axi`; it does not protect itself from a malicious master on `s_axi_ctrl`. If your threat model includes the latter, place the control port behind something that arbitrates access to it — a second firewall instance is one option.

A one-way lock bit is a natural addition and is not currently implemented.

9.6 Combinational Rule Lookup

The lookup is a combinational priority chain over `NUM_RULES` entries, so it is the expected critical path and it scales with `NUM_RULES`. With no synthesis data there is no guidance on where it stops meeting timing. If it limits f_{MAX} , register the lookup result and accept one more cycle in EVAL.

9.7 Shared Fault Capture

`FAULT_ADDR` and `FAULT_INFO` are single registers shared by both datapaths. Simultaneous read and write faults both set their sticky `STATUS` bits correctly, but only one address and one type are captured — and the two are resolved by *different* rules:

Table 40. Fault Capture Resolution

Field	Resolved by
<code>FAULT_ADDR</code>	Write side wins
<code>FAULT_INFO.WAS_WRITE</code>	Write side wins
<code>FAULT_INFO.FAULT_TYPE</code>	Type precedence across both datapaths: <code>TIMEOUT</code> > <code>PERM</code> > <code>ADDR</code>

The two rules agree whenever both datapaths fault with the same type, or when only one faults. They disagree when the datapaths fault in the same cycle with different types: the address and direction then describe the write while the type may describe the read.

This is a consequence of the capture being one register pair fed by three independent fault sources, and it is not currently arbitrated. Software that must attribute every fault individually needs to correlate with its own transaction log. Giving each datapath its own capture pair would remove the ambiguity and is a candidate for a future revision.

9.8 Synchronous Reset Only

`resetn` is sampled synchronously. If your reset source is asynchronous, put a reset synchroniser ahead of the core. Platform Designer's reset bridge does this for you.

10. Document Revision History

Table 41. Document Revision History

Document version	Core version	Date	Changes
1.2	2.0	August 2026	Recorded hardware verification on a Terasic DE10-Lite: both example designs run on a physical board (Section 8.4), the Platform Designer component flow works, and the core is now measured at a second ADDR_WIDTH (Section 1.3). Several rows moved out of "not verified". No RTL change.
1.1	2.0	August 2026	Added measured resource and f_{MAX} figures (Section 1.3), which were previously uncharacterised. Corrected the recovery sequence: the protected peripheral's reset must remain asserted while RECOVERY.UNBLOCK is written, and the sequence is now seven steps rather than six (Sections 3.5 and 7.4). Added the hardware demonstration (Section 8.4) and updated what has not been verified. No RTL change.
1.0	2.0	August 2026	Initial release of this user guide

Table 42. Core Revision History

Core version	CORE_INFO	Changes
2.0	0x0200	Removed the m_axi_resetrn output and the RESET_HOLD_CYCLES parameter. Added the RECOVERY register with UNBLOCK, and STATUS bits BLOCKED, WR/RD_RESP_BUSY and WR/RD_CMD_STUCK. Recovery became an explicit software sequence. Sources converted to SystemVerilog. Breaking change.
1.2	0x0102	Fixed two AXI compliance defects on the control port: AWREADY / WREADY were asserted while a BVALID was outstanding, and ARREADY while an RVALID was, so a pipelining master lost responses. Fixed a rule-register byte-merge defect when ADDR_WIDTH < 32. Removed dead discard-pending logic.
1.1	0x0101	Split the single access-violation assertion into separate read and write assertions, which exposed that no test had ever denied a read.
1.0	0x0100	Initial release